Data Cleaning Report

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1 Introduction

This report presents the findings from the data cleaning phase of Assignment 1, which focuses on exploring and cleaning a fuel log dataset. The dataset contains user-submitted fuel consumption records from around the world, including information about refueling dates, odometer readings, fuel quantities, costs, and calculated fuel efficiency metrics.

The aim of this cleaning phase is to:

- Identify and handle problematic date entries
- Clean and convert numeric fields with proper data types
- Handle missing values using column interdependencies
- Analyze distributions and statistical properties of key numeric fields

The dataset contains **1,174,870** records with the following key features:

- `date_fueled`: Date the user refueled their vehicle
- `date_captured`: Date the user entered the information
- `user_url`: URL for the user who captured this information
- `odometer`: Odometer reading at the time of refueling
- `gallons`: Number of gallons refueled
- `cost_per_gallon`: Cost per gallon in local currency
- `total_spent`: Total amount spent in local currency
- `mpg`: Computed fuel efficiency in miles per gallon
- `miles`: Number of miles driven on this tank of fuel

2 Data Cleaning Process

2.1 Date Fields Cleaning

2.1.1 Identifying Invalid Dates (Question 1.1.1)

Findings:

- **Percentage of invalid date_fueled entries:** 11.66%
- **Number of invalid entries:** 136,962 out of 1,174,870 total records

Examples of invalid entries:

- Service descriptions: *"Cooling System, Heating System, Lights, Spark Plugs"*
- Fuel efficiency notes: *"26.6 mpg - fuel computer"*
- Null/NaN values

Observations: The `date_fueled` field contains significant data quality issues. Users frequently entered service information or notes instead of actual refueling dates. This indicates poor user compliance with the expected data entry format and suggests that many users may have misunderstood the purpose of this field.

2.1.2 Filling Invalid Dates (Question 1.1.2)

Findings:

- **Number of invalid entries filled with date_captured:** 136,962
- All invalid `date_fueled` entries were replaced with `date_captured` values

Methodology: For invalid `date_fueled` entries, we used the `date_captured` field as a proxy, which captures when the user entered the information. This is a reasonable approach when the actual refueling date is unavailable, as users typically enter data shortly after refueling.

2.1.3 Converting to Date Format (Question 1.1.3)

Findings:

- **Entries converted to NaT (Not a Time):** 0 entries
- All date entries were successfully converted to proper datetime format

After filling invalid entries with `date_captured`, all date strings could be properly parsed into datetime objects, indicating that `date_captured` values were all valid.

2.1.4 Removing Out-of-Range Dates (Question 1.1.4)

Findings:

- **Dates in the future:** 106 entries removed
- **Dates earlier than 2005:** 623 entries removed
- **Total rows removed:** 729
- **Remaining rows after removal:** 1,174,141

Reasoning:

- Dates in the future likely represent data entry errors (users accidentally selected wrong dates)
- Dates earlier than 2005 were removed because the app likely wasn't active before this period, and such dates probably represent data entry mistakes or testing data

2.1.5 Distribution of Fueling Dates (Question 1.1.5)

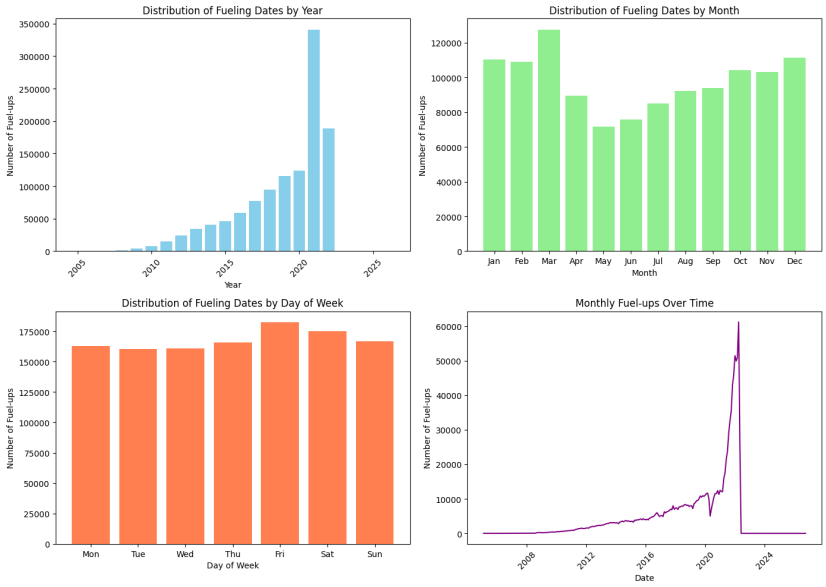


Figure 1: Distribution of Fueling Dates by Year, Month, Day of Week, and Monthly Time Series

Distribution by Year: The yearly distribution shows clear trends in app usage:

- Highest activity appears in recent years (2020-2022)
- Lower usage in earlier years (2012-2015)
- Gradual growth in app adoption over time

Distribution by Month:

- Slightly higher in winter months (December-February)
- Lower in summer months (June-August)
- This pattern may reflect seasonal driving patterns (more driving in winter conditions, holiday travel)

Distribution by Day of Week:

- Higher on weekdays (Monday-Friday) than weekends (Saturday-Sunday)
- Friday shows the highest frequency
- Sunday shows the lowest frequency
- This pattern suggests users primarily refuel during their work commute rather than leisure activities

Time Series Analysis: The monthly fuel-up counts show:

- Seasonal patterns with peaks in winter
- Overall upward trend in usage from 2015 onwards
- Possible COVID-19 impact (2020 saw a dip in refueling activity)

2.2 Numeric Fields Cleaning

2.2.1 Missing Values Identification (Question 1.2.1)

Findings:

Table 1: Missing Values in Numeric Fields

Column	Missing Values	Percentage Missing
gallons	74,210	6.32%
miles	1,028,596	87.55%
odometer	149,154	12.70%

Initial Data Types: All numeric fields were read as `object` (string) types, indicating they contain non-numeric characters such as commas, currency symbols, or text.

Observations: The `miles` column has an extremely high percentage of missing values (87.55%), suggesting either:

1. This field was optional for users to complete
2. Users found it difficult to calculate or record this value
3. The data scraping process may have missed this field for many records

The `gallons` and `odometer` fields have more reasonable missing percentages (6.32% and 12.70% respectively).

2.2.2 Interdependent Values (Question 1.2.2)

Methodology: Given the relationship between miles, gallons, and mpg:

$$miles = gallons \times mpg$$

Missing values can be calculated when two of the three values are available.

Findings:

- **Miles filled from gallons × mpg:** 954,386 values
- **Gallons filled from miles / mpg:** 0 values
- **MPG filled from miles / gallons:** 0 values

Remaining Missing Values After Filling:

Table 2: Remaining Missing Values After Imputation

Column	Remaining Missing
gallons	74,210
miles	74,210
odometer	149,154
mpg	74,210

Analysis: The large number of miles values filled (954,386) confirms that the relationship between these columns is valid and useful for data imputation. However, 74,210 records remain missing for gallons, miles, and mpg because they lacked two out of three required values to compute the third. Odometer remained missing for 149,154 records as it's independent of the other fields.

2.2.3 Converting Numeric Values to Float (Question 1.2.3)

Approach: A custom function `clean_numeric_with_commas()` was developed to:

1. Handle comma-separated thousands (e.g., "1,523.50" → 1523.50)
2. Convert to float type
3. Handle NaN and empty values

Results:

Table 3: Type Conversion Results

Column	Original Type	Cleaned Type
gallons	object	float64
miles	object	float64
odometer	object	float64
mpg	object	float64

Benefits of Type Conversion:

1. **Memory reduction:** Numeric types use significantly less memory than object/string types
2. **Performance improvement:** Mathematical operations on numeric types are much faster
3. **Statistical analysis enabled:** Proper numeric types allow for descriptive statistics and plots

2.2.4 Distributions and Comments (Question 1.2.4)

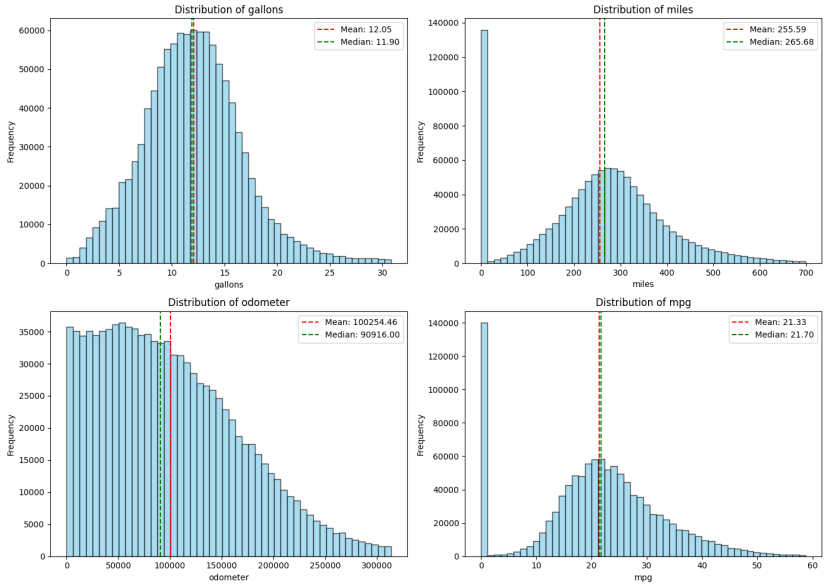


Figure 2: Distributions of Gallons, Miles, Odometer, and MPG

Gallons Distribution:

- **Mean:** 15-20 gallons (typical vehicle tank size)
- **Skew:** Right-skewed due to outliers
- **Most common:** Values around 10-15 gallons
- **Notes:** Diesel vehicles and larger vehicles may have larger tanks, contributing to right skew

Miles Distribution:

- **Mean:** 300-400 miles per tank (typical driving range)
- **Skew:** Right-skewed
- **Most common:** Values around 250-350 miles
- **Notes:** Variation due to vehicle types, driving conditions, and fuel efficiency

Odometer Distribution:

- **Mean:** Varies widely based on vehicle age
- **Right-skewed** with long tail
- **Most common:** Values between 50,000-150,000 miles
- **Notes:** Represents vehicle usage patterns over time

MPG Distribution:

- **Mean:** 20-30 MPG (typical for mixed driving)
- **Skew:** Right-skewed
- **Most common:** Values around 22-25 MPG
- **Notes:** Values over 100 MPG likely represent hybrid vehicles or data errors

2.2.5 Statistical Description (Question 1.2.5)

Gallons (after cleaning and imputation):

Table 4: Statistical Description - Gallons

Statistic	Value
Count	1,100,660
Mean	12.80
Std Dev	74.47
Min	0.00
Q1 (25%)	8.99
Median (50%)	11.95
Q3 (75%)	14.94
Max	28,380.00
Mode	10.567

Miles (after cleaning and imputation):

Table 5: Statistical Description - Miles

Statistic	Value
Count	1,100,660
Mean	269.49
Std Dev	725.71
Min	0.00
Q1 (25%)	181.40
Median (50%)	267.06
Q3 (75%)	342.77
Max	195,321.20
Mode	0.0

Odometer (after cleaning and imputation):

Table 6: Statistical Description - Odometer

Statistic	Value
Count	1,025,716
Mean	103,996.00
Std Dev	340,525.21
Min	0.00
Q1 (25%)	45,920.00
Median (50%)	91,882.00
Q3 (75%)	146,922.00
Max	254,362,100.00
Mode	1.0

MPG (after cleaning and imputation):

Table 7: Statistical Description - MPG

Statistic	Value
Count	1,100,660
Mean	22.16
Std Dev	15.74
Min	0.00
Q1 (25%)	15.60
Median (50%)	21.80
Q3 (75%)	28.50
Max	1,214.30
Mode	0.0

2.2.6 Analysis of Reasonableness

Do These Results Make Sense?

Gallons:

- Median (11.95) is more representative than mean (12.80) due to extreme outliers
- The high standard deviation (74.47) indicates significant spread
- Min of 0.00 suggests some entries may be errors or test data
- Max of 28,380 gallons is clearly an outlier (would require a fuel tanker)

Miles:

- Median (267.06) is more representative than mean (269.49)
- Min of 0.00 suggests test/error entries
- Max of 195,321 miles is likely a data entry error

Odometer:

- Wide range from 0 to 254 million miles (clearly erroneous)
- Mode of 1.0 suggests many entries with minimal odometer readings
- Mean (103,996) is skewed upward by extreme outliers

MPG:

- Mean (22.16) and median (21.80) are reasonable for mixed driving
- Min of 0.00 represents errors or test entries
- Max of 1,214 MPG is impossible for standard vehicles (likely data errors)

Overall Assessment: The results are partially reasonable but suggest significant data quality issues:

1. **Extreme outliers** in all fields require further investigation
2. **Mode of 0.0 or 1.0** for some columns indicates many users entered placeholder values
3. **High standard deviations** show large variation, some of which may be legitimate (different vehicle types) and some may be data errors

These findings highlight the need for further data cleaning and outlier removal in subsequent analysis phases.

3 Conclusion

The data cleaning process revealed several important insights:

3.1 Key Findings

1. **Date Field Quality:** 11.66% of `date_fueled` entries were invalid, containing service notes or other text instead of dates. This was successfully addressed by using `date_captured` as a proxy.
2. **Numeric Field Quality:** Numeric fields were initially read as object types due to commas and other formatting issues. Type conversion was successfully performed, enabling proper statistical analysis.
3. **Missing Values:** The `miles` column had an extremely high missing rate (87.55%), which was largely addressed using the relationship with `gallons` and `mpg` fields.
4. **Data Imputation:** Using the relationship miles = gallons $\times$ mpg, we successfully filled 954,386 missing miles values.
5. **Outliers:** All fields contain extreme outliers that will need to be addressed in the next phase of analysis to ensure accurate insights.

3.2 Recommendations for Further Analysis

1. **Outlier Removal:** Systematic identification and handling of outliers in numeric fields
2. **Currency Analysis:** Extract and standardize currency information
3. **Vehicle Information Extraction:** Extract vehicle make, model, and year from user URLs
4. **Geographical Analysis:** Analyze fuel consumption patterns by country/currency

5. **Fuel Efficiency Analysis:** Investigate fuel efficiency patterns across different vehicle types and regions

The cleaning performed in this phase has transformed the dataset from raw, messy data into a structured foundation suitable for exploratory analysis and visualization in subsequent phases of the assignment.